DATABASE SYSTEMS

Laboratory Work #5

Constraints, Validation Rules and Default Values

DBMS	PostgreSQL 18
Client	pgAdmin 4
Database	DB26_Labs
Report language	English

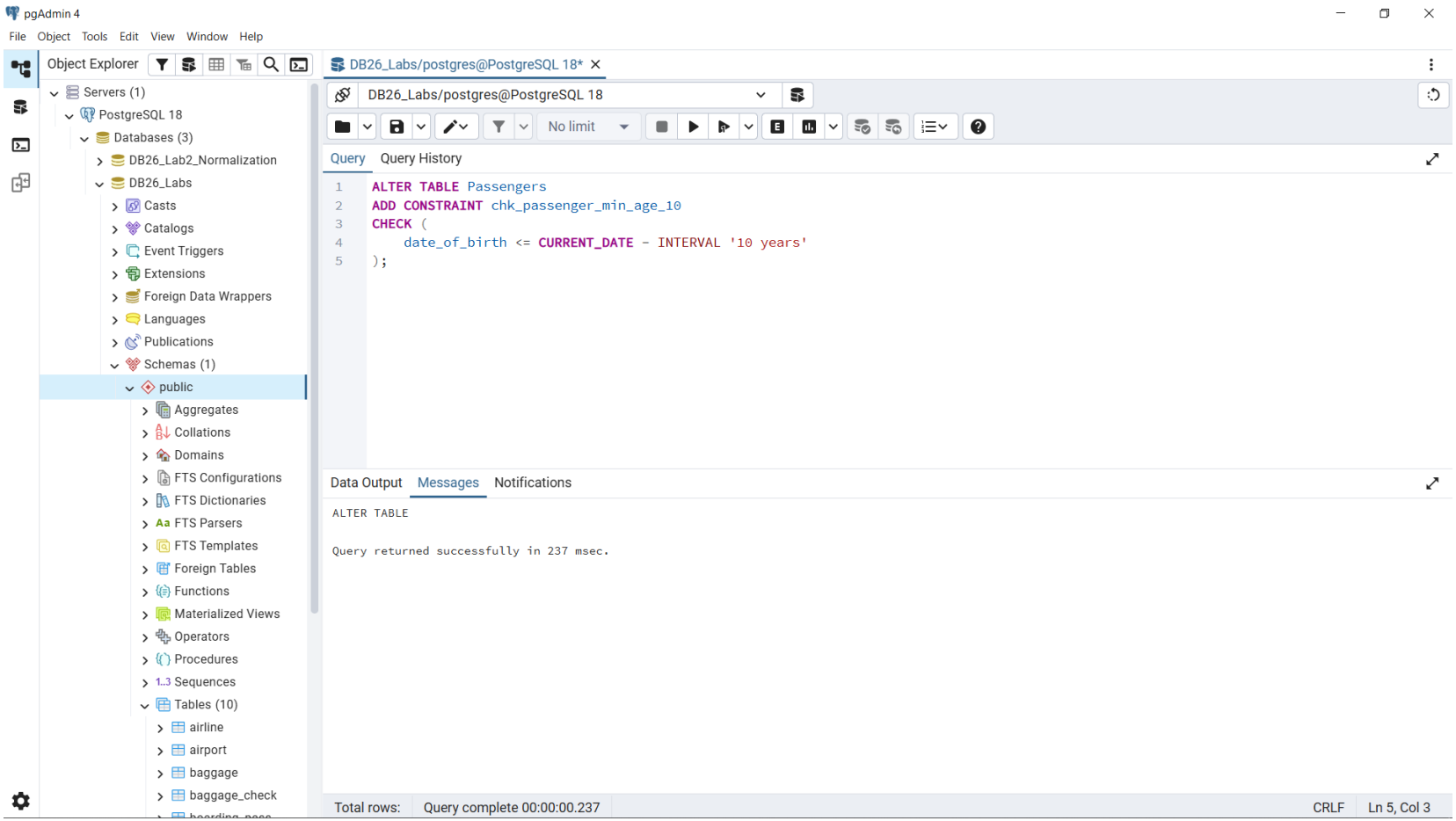
Objective

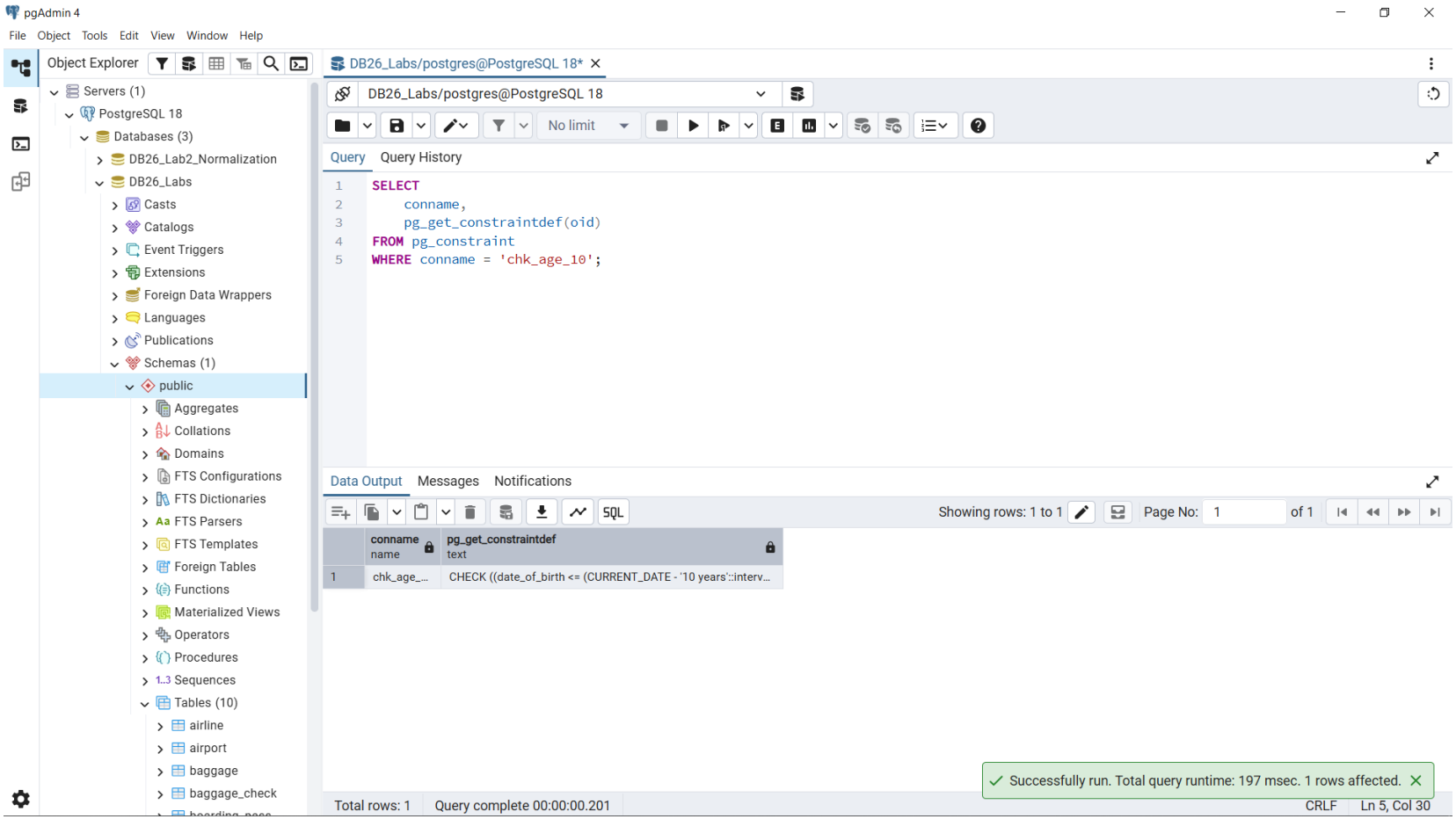
To practice integrity constraints and default values in PostgreSQL by applying CHECK and UNIQUE constraints, conditional validation rules, calculated ticket discounts, and automatic default values to the airport database from previous laboratory works.

Task 1

Add a CHECK constraint so that every passenger is at least 10 years old.

```
ALTER TABLE Passengers
ADD CONSTRAINT chk_passenger_min_age_10
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
```

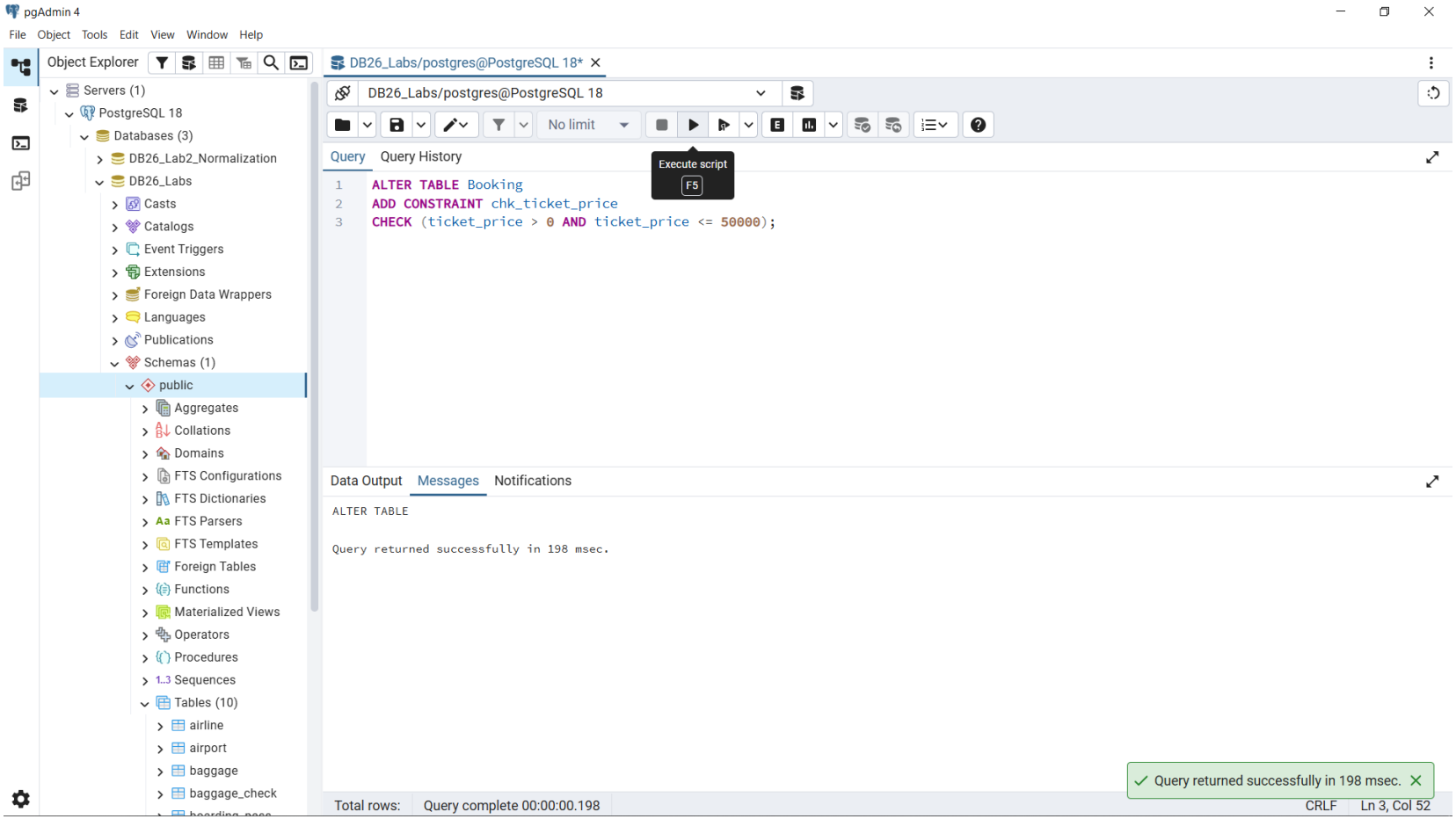


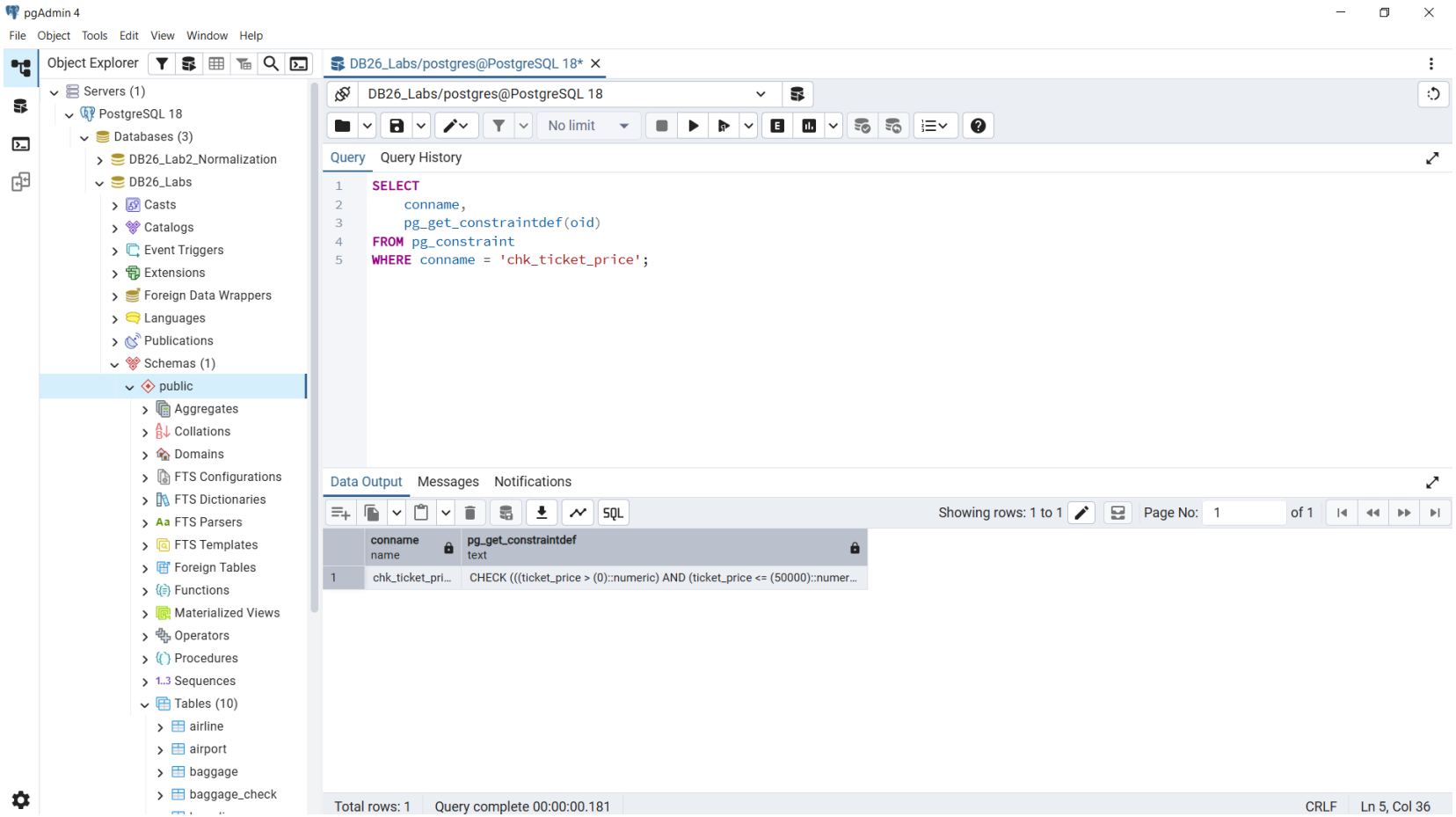

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Query
Query History
1 SELECT
2 conname,
3 pg_get_constraintdef(oid)
4 FROM pg_constraint
5 WHERE conname = 'chk_age_10';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
1
chk_age_...
CHECK ((date_of_birth <= (CURRENT_DATE - '10 years'::interv...
Successfully run. Total query runtime: 197 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.201
CRLF
Ln 5, Col 30

Task 2

Add a CHECK constraint for booking price. The implementation uses the logical range 0 < ticket_price <= 50000 because the assignment text contains an apparent contradiction.

```
ALTER TABLE Booking
ADD CONSTRAINT chk_ticket_price
CHECK (ticket_price > 0 AND ticket_price <= 50000);
```

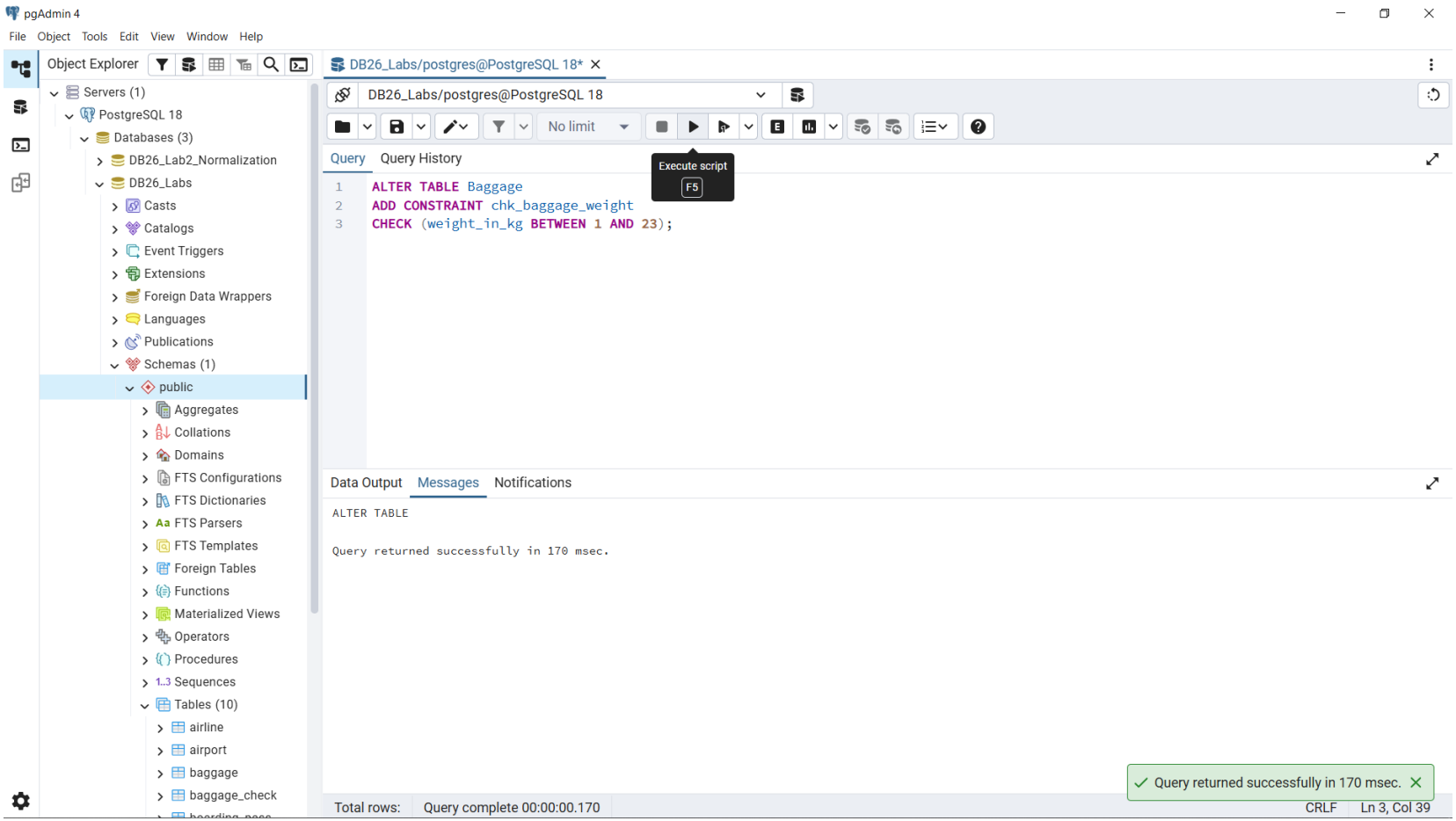


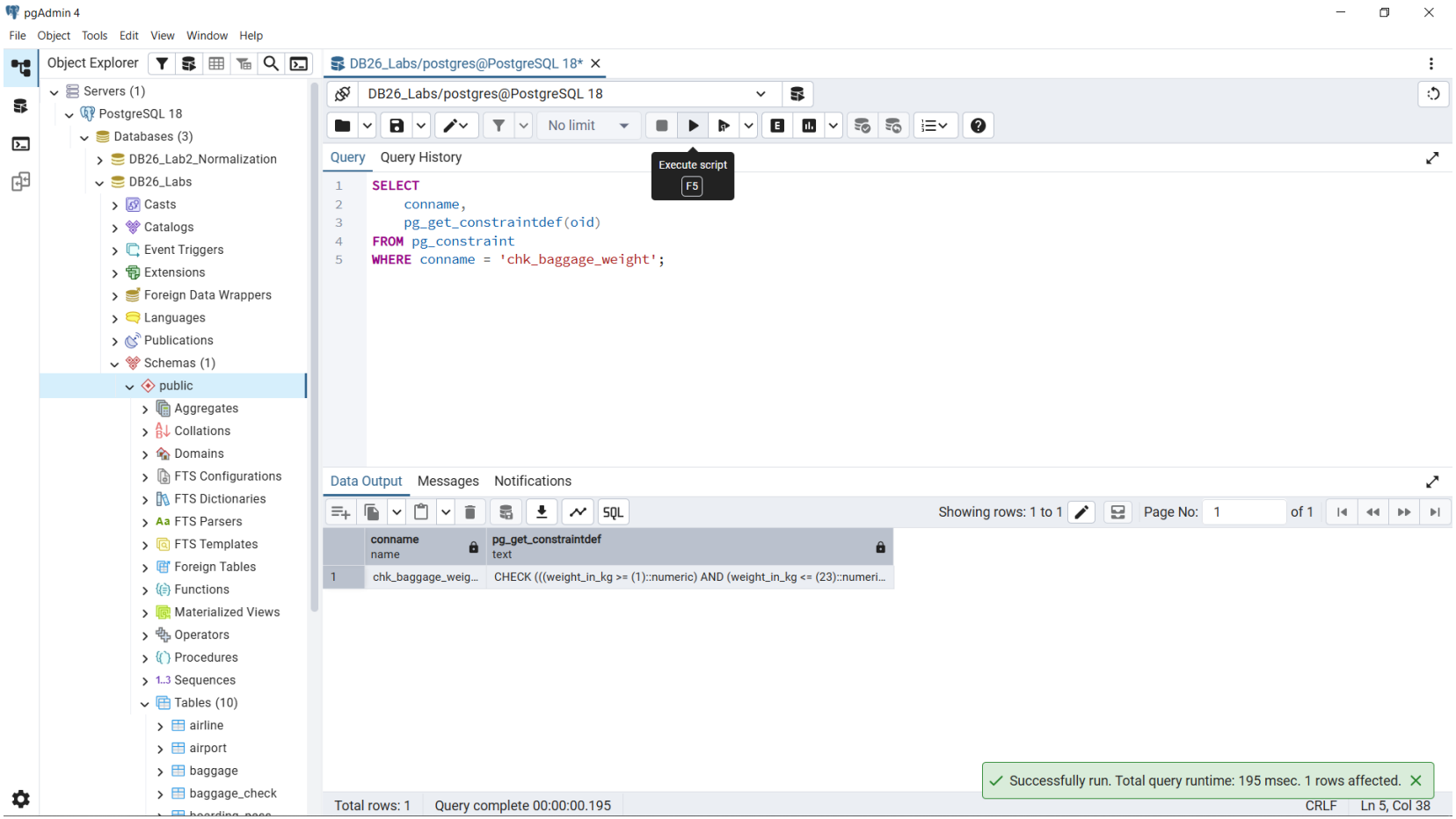

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE conname = 'chk_ticket_price';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
1
chk_ticket_pri...
CHECK (((ticket_price > (0)::numeric) AND (ticket_price <= (50000)::numer...
Total rows: 1
Query complete 00:00:00.181
CRLF
Ln 5, Col 36

Task 3

Add a CHECK constraint that allows baggage weight only between 1 and 23 kg.

```
ALTER TABLE Baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
```

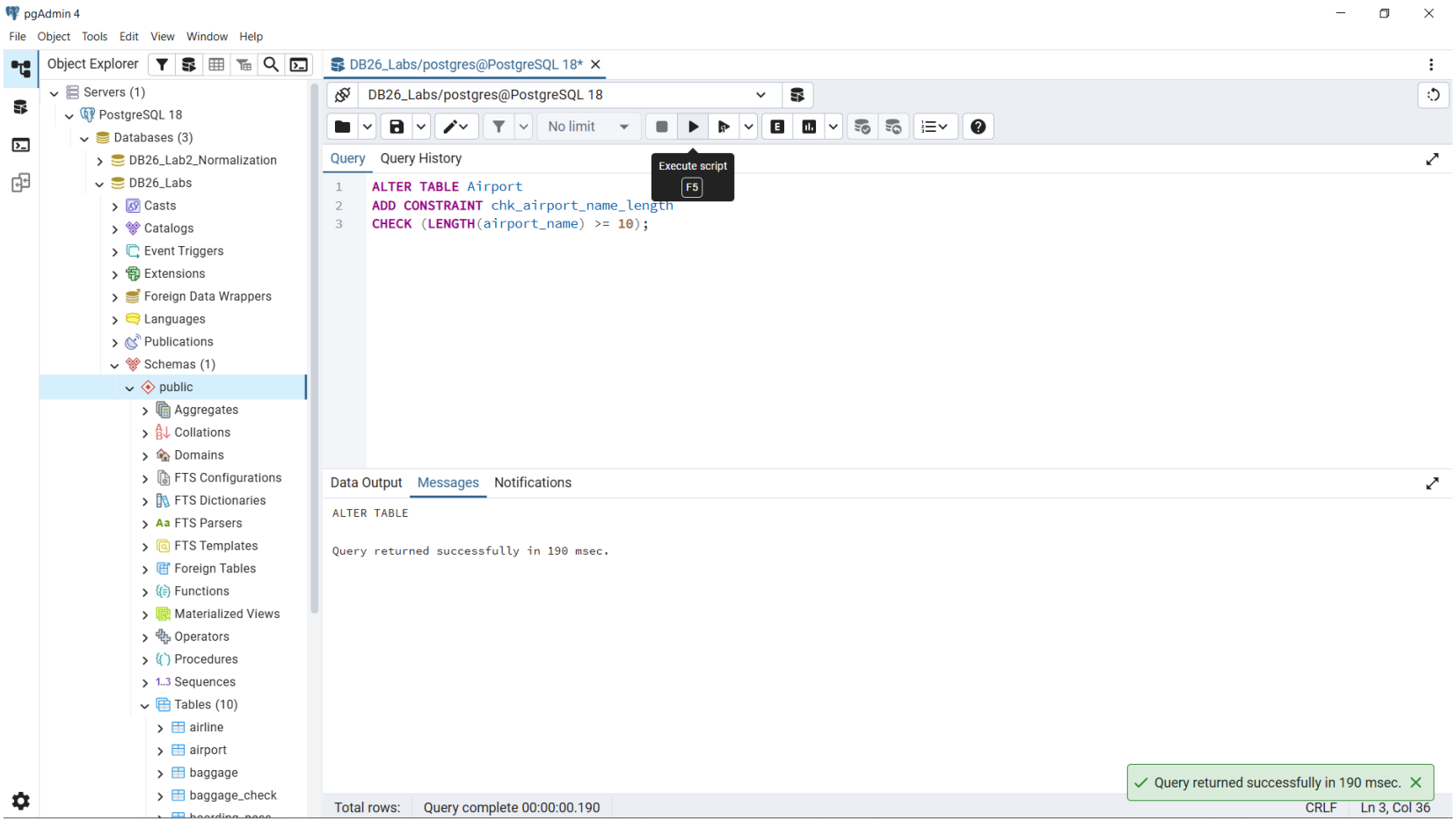


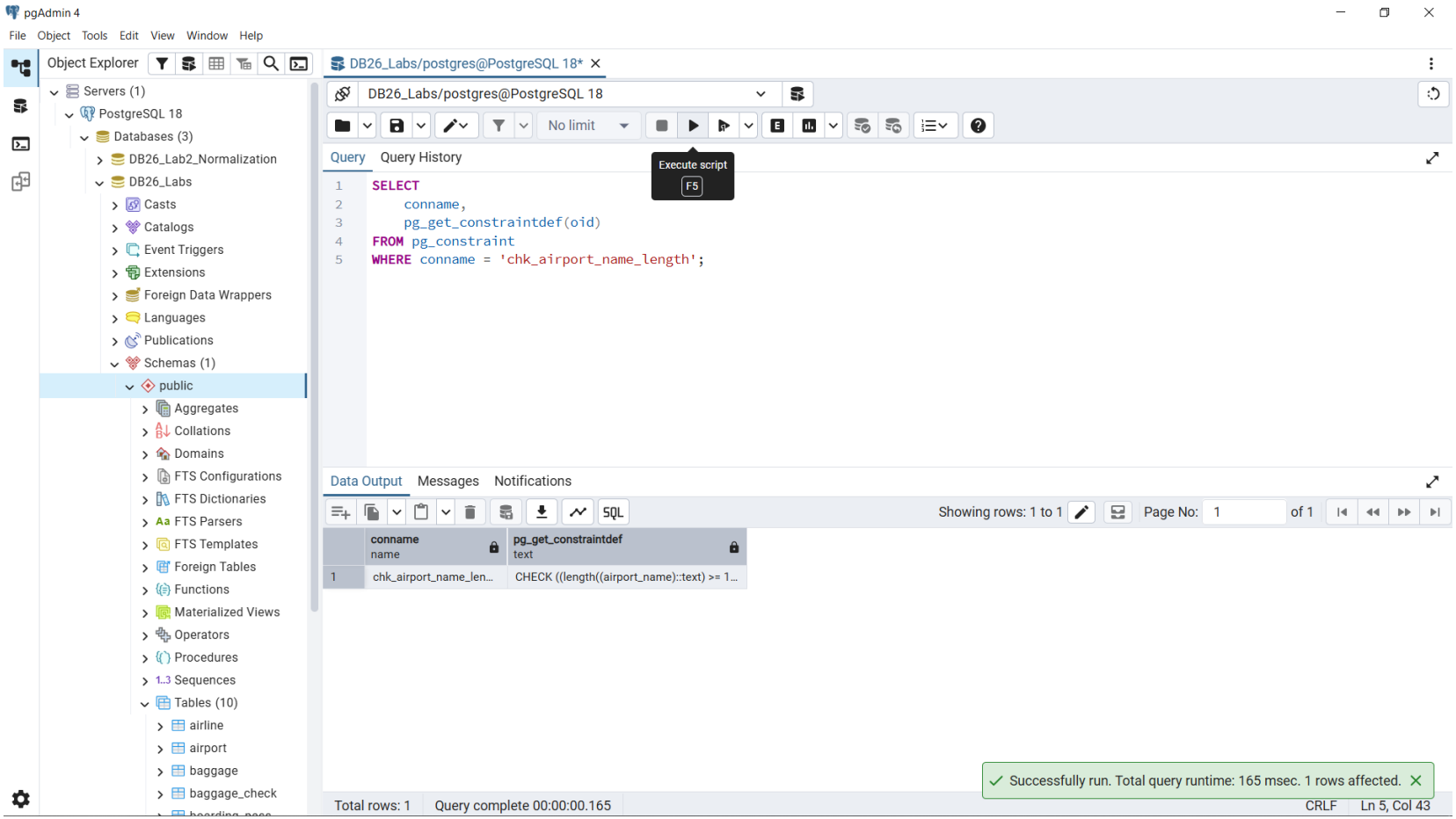

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE conname = 'chk_baggage_weight';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
1
chk_baggage_weig...
CHECK (((weight_in_kg >= (1)::numeric) AND (weight_in_kg <= (23)::numeri...
Successfully run. Total query runtime: 195 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.195
CRLF
Ln 5, Col 38

Task 4

Add a CHECK constraint requiring airport names to contain at least 10 characters.

```
ALTER TABLE Airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
```

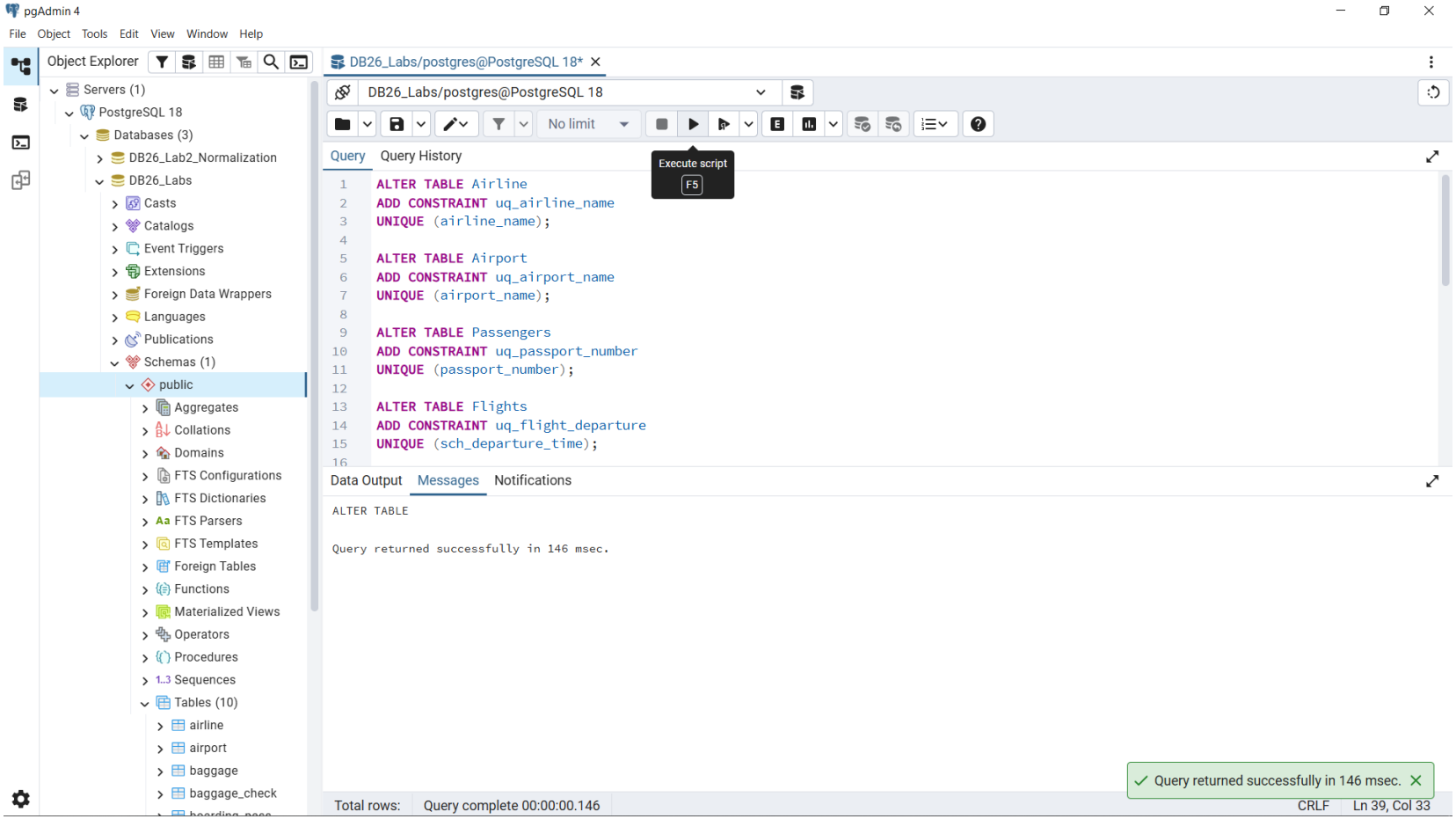


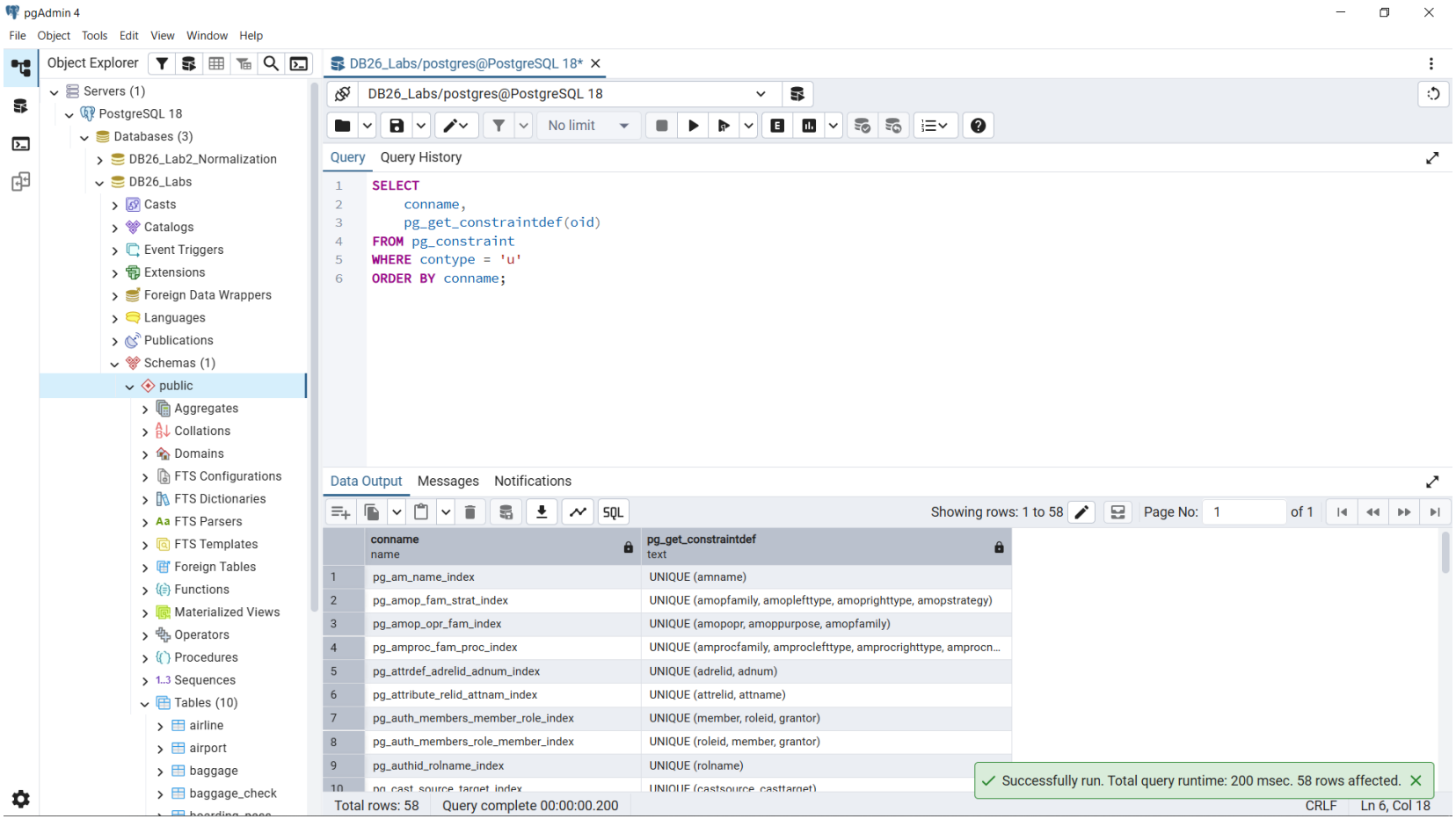

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE conname = 'chk_airport_name_length';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
1
chk_airport_name_len...
CHECK ((length((airport_name)::text) >= 1...
Successfully run. Total query runtime: 165 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.165
CRLF
Ln 5, Col 43

Task 5

Add UNIQUE constraints to suitable columns in the database tables.

```
ALTER TABLE Airline ADD CONSTRAINT uq_airline_name UNIQUE (airline_name);
ALTER TABLE Airport ADD CONSTRAINT uq_airport_name UNIQUE (airport_name);
ALTER TABLE Passengers ADD CONSTRAINT uq_passport_number UNIQUE (passport_number);
ALTER TABLE Flights ADD CONSTRAINT uq_flight_departure UNIQUE (sch_departure_time);
-- Additional UNIQUE constraints were also added to the remaining tables.
```

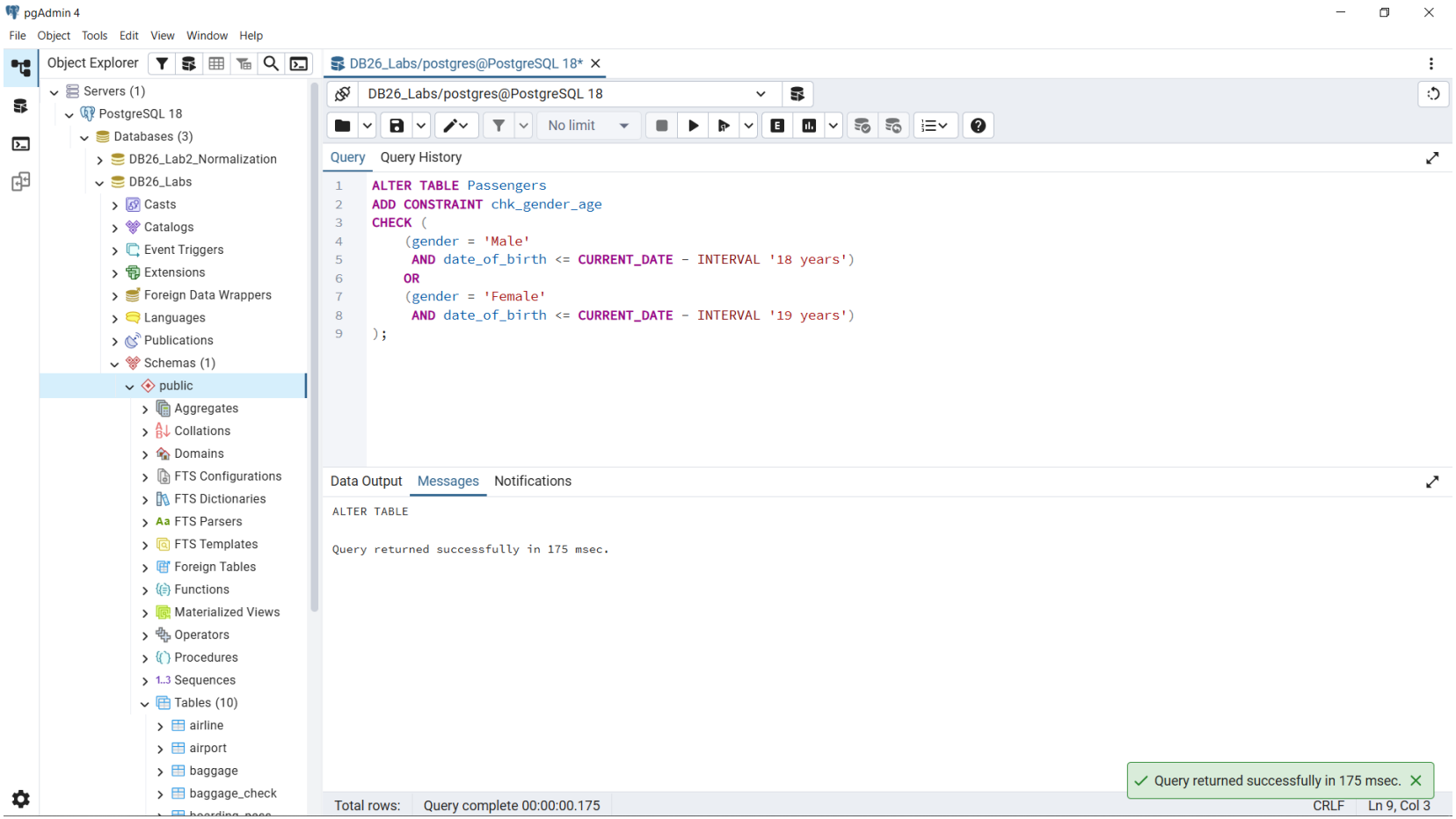


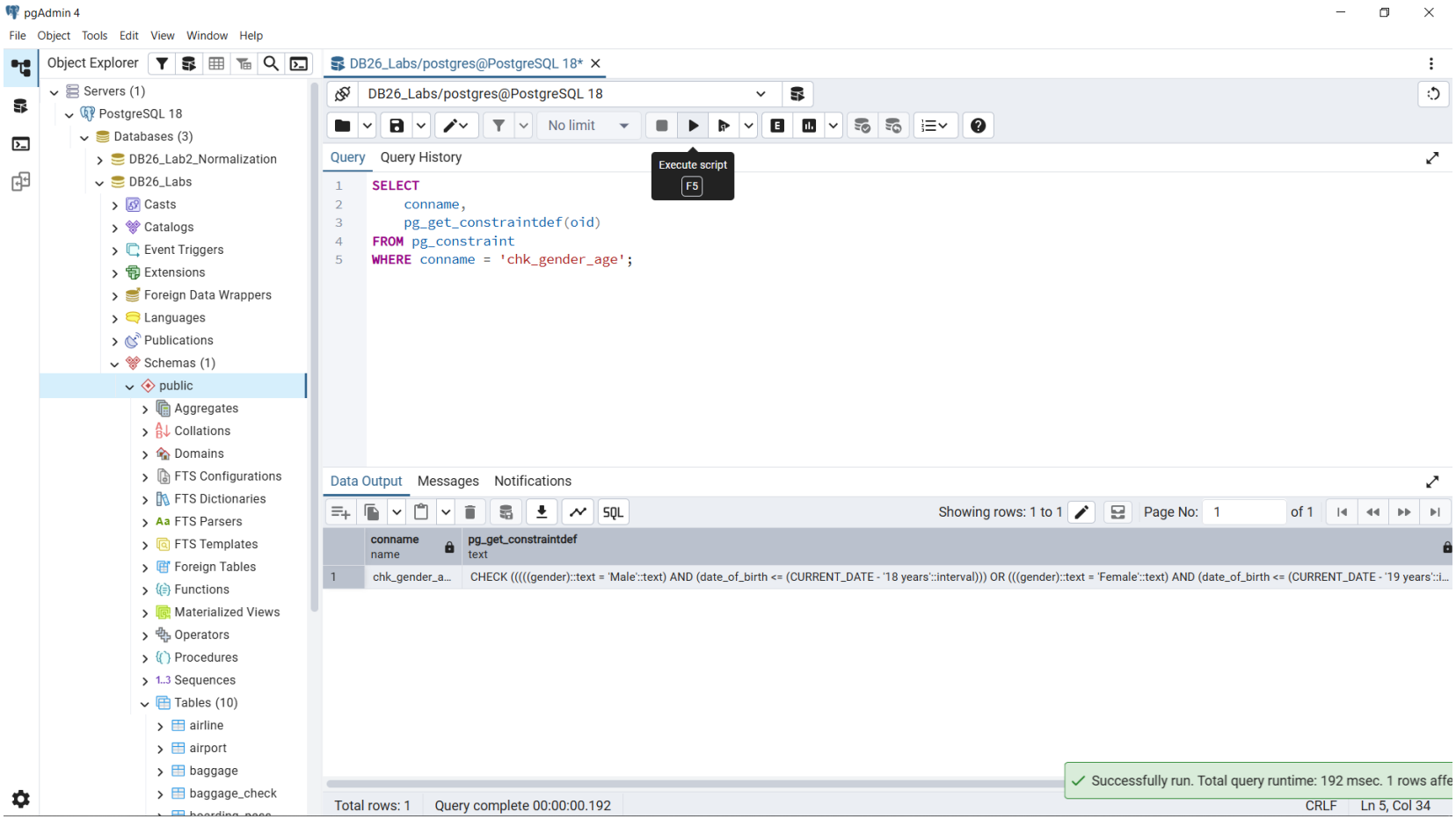

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE contype = 'u'
ORDER BY conname;
Data Output
Messages
Notifications
Showing rows: 1 to 58
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
1 pg_am_name_index UNIQUE (amname)
2 pg_amop_fam_strat_index UNIQUE (amopfamily, amoplefttype, amoprighttype, amopstrategy)
3 pg_amop_opr_fam_index UNIQUE (amopopr, amoppurpose, amopfamily)
4 pg_amproc_fam_proc_index UNIQUE (amprocfamily, amproclefttype, amprocrighttype, amprocn...
5 pg_attrdef_adrelid_adnum_index UNIQUE (adrelid, adnum)
6 pg_attribute_relid_attnam_index UNIQUE (attrelid, attname)
7 pg_auth_members_member_role_index UNIQUE (member, roleid, grantor)
8 pg_auth_members_role_member_index UNIQUE (roleid, member, grantor)
9 pg_authid_rolname_index UNIQUE (rolname)
Successfully run. Total query runtime: 200 msec. 58 rows affected.
Total rows: 58
Query complete 00:00:00.200
CRLF
Ln 6, Col 18

Task 6

Add a CHECK constraint: male passengers must be at least 18 years old and female passengers at least 19 years old.

```
ALTER TABLE Passengers
ADD CONSTRAINT chk_gender_age
CHECK (
  (gender = 'Male' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
  OR
  (gender = 'Female' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
```

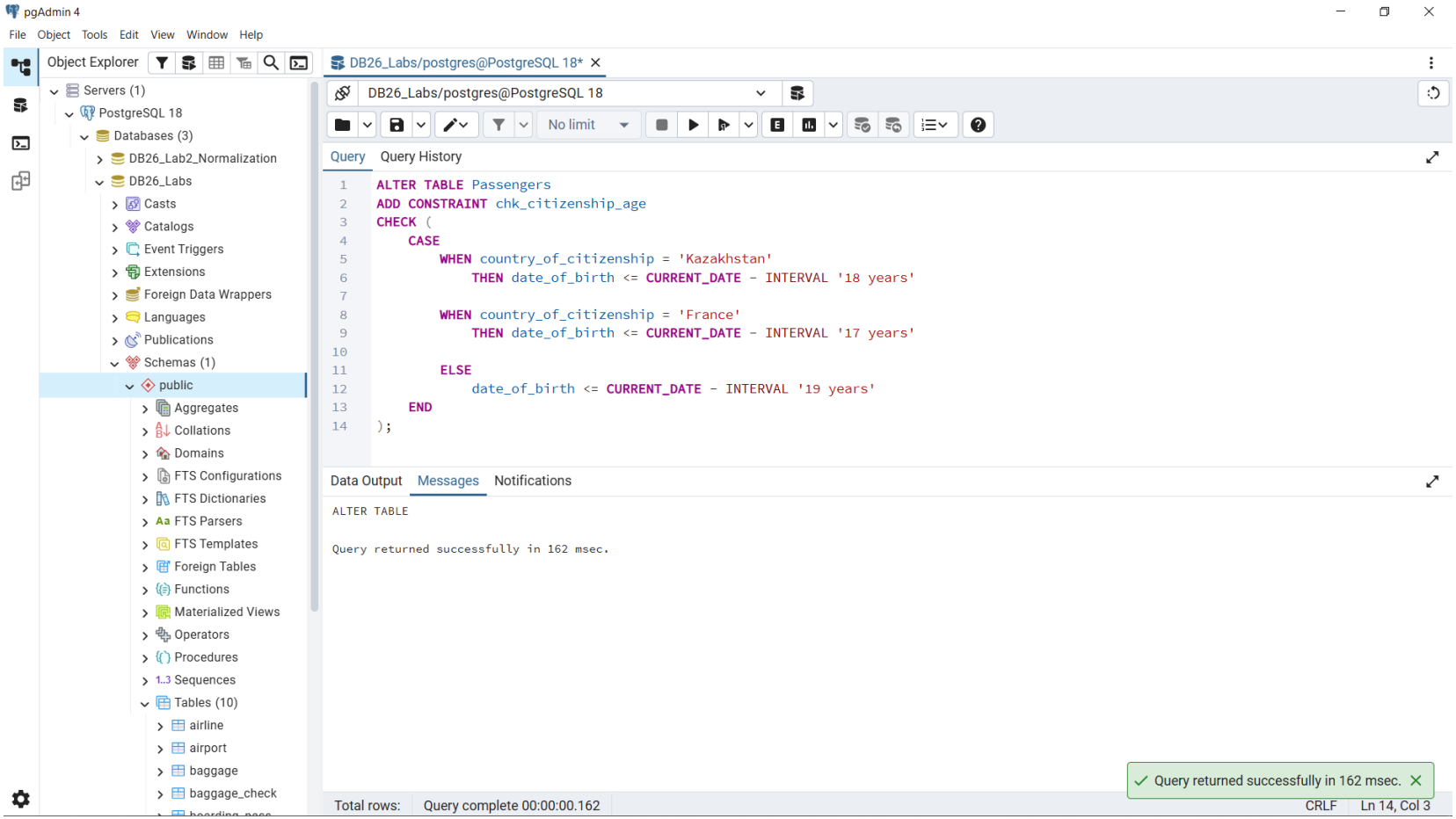


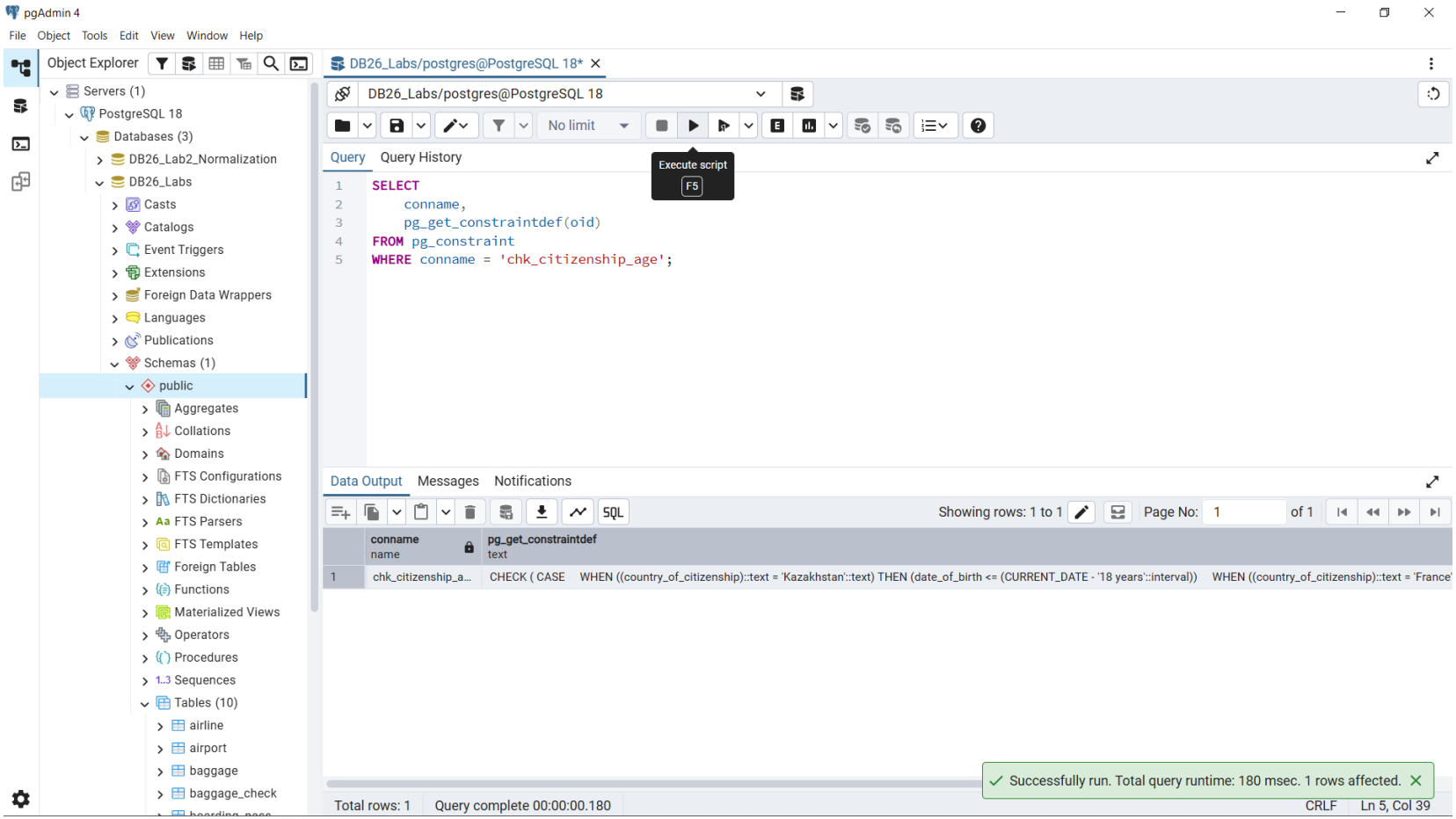

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE conname = 'chk_gender_age';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
chk_gender_a...
CHECK (((((gender)::text = 'Male'::text) AND (date_of_birth <= (CURRENT_DATE - '18 years'::interval))) OR (((gender)::text = 'Female'::text) AND (date_of_birth <= (CURRENT_DATE - '19 years'::i...
Successfully run. Total query runtime: 192 msec. 1 rows affe
Total rows: 1
Query complete 00:00:00.192
CRLF
Ln 5, Col 34

Task 7

Add age rules based on country_of_citizenship: Kazakhstan >= 18, France >= 17, and all other countries >= 19.

```
ALTER TABLE Passengers
ADD CONSTRAINT chk_citizenship_age
CHECK (
  CASE
    WHEN country_of_citizenship = 'Kazakhstan' THEN date_of_birth <= CURRENT_DATE - INTERVAL '18 years'
    WHEN country_of_citizenship = 'France' THEN date_of_birth <= CURRENT_DATE - INTERVAL '17 years'
    ELSE date_of_birth <= CURRENT_DATE - INTERVAL '19 years'
  END
);
```




pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
SELECT
conname,
pg_get_constraintdef(oid)
FROM pg_constraint
WHERE conname = 'chk_citizenship_age';
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
conname
name
pg_get_constraintdef
text
chk_citizenship_a...
CHECK (CASE WHEN ((country_of_citizenship)::text = 'Kazakhstan'::text) THEN (date_of_birth <= (CURRENT_DATE - '18 years'::interval)) WHEN ((country_of_citizenship)::text = 'France'
Successfully run. Total query runtime: 180 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.180
CRLF
Ln 5, Col 39

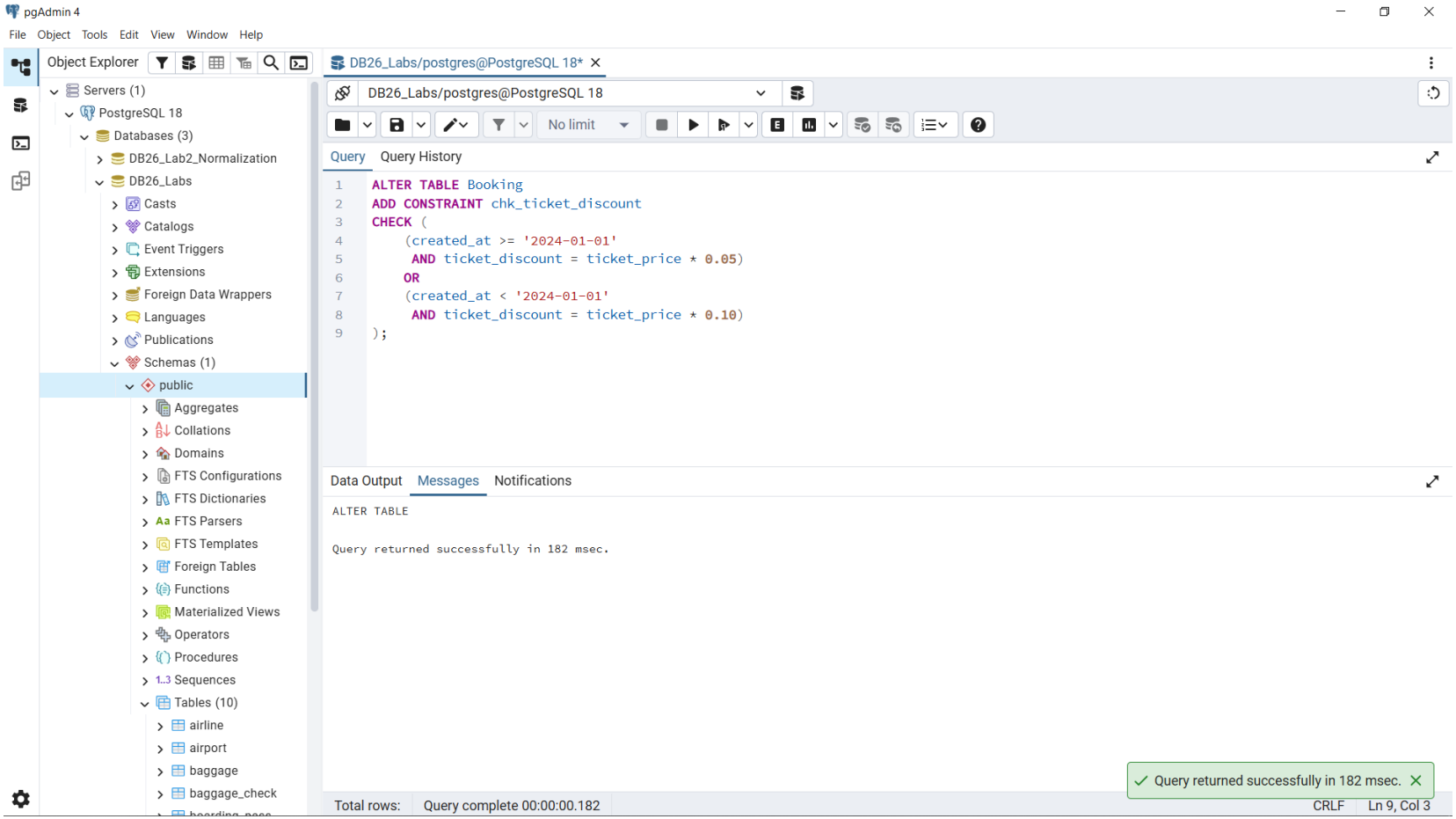
Task 8

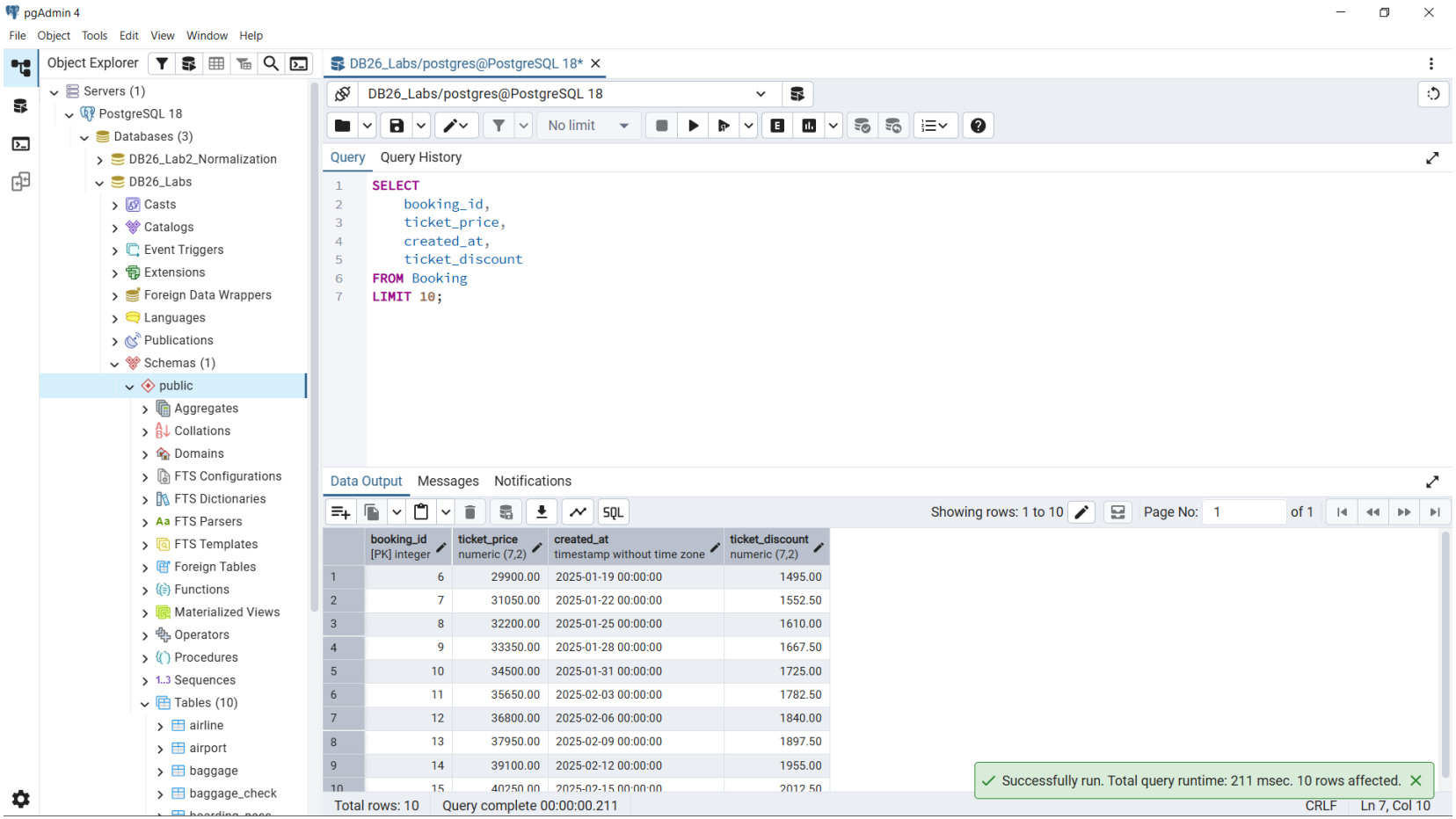
Add ticket_discount to Booking and validate a 5% discount for bookings created on/after 2024-01-01 and a 10% discount for earlier bookings.

```
ALTER TABLE Booking ADD COLUMN ticket_discount DECIMAL(7,2);

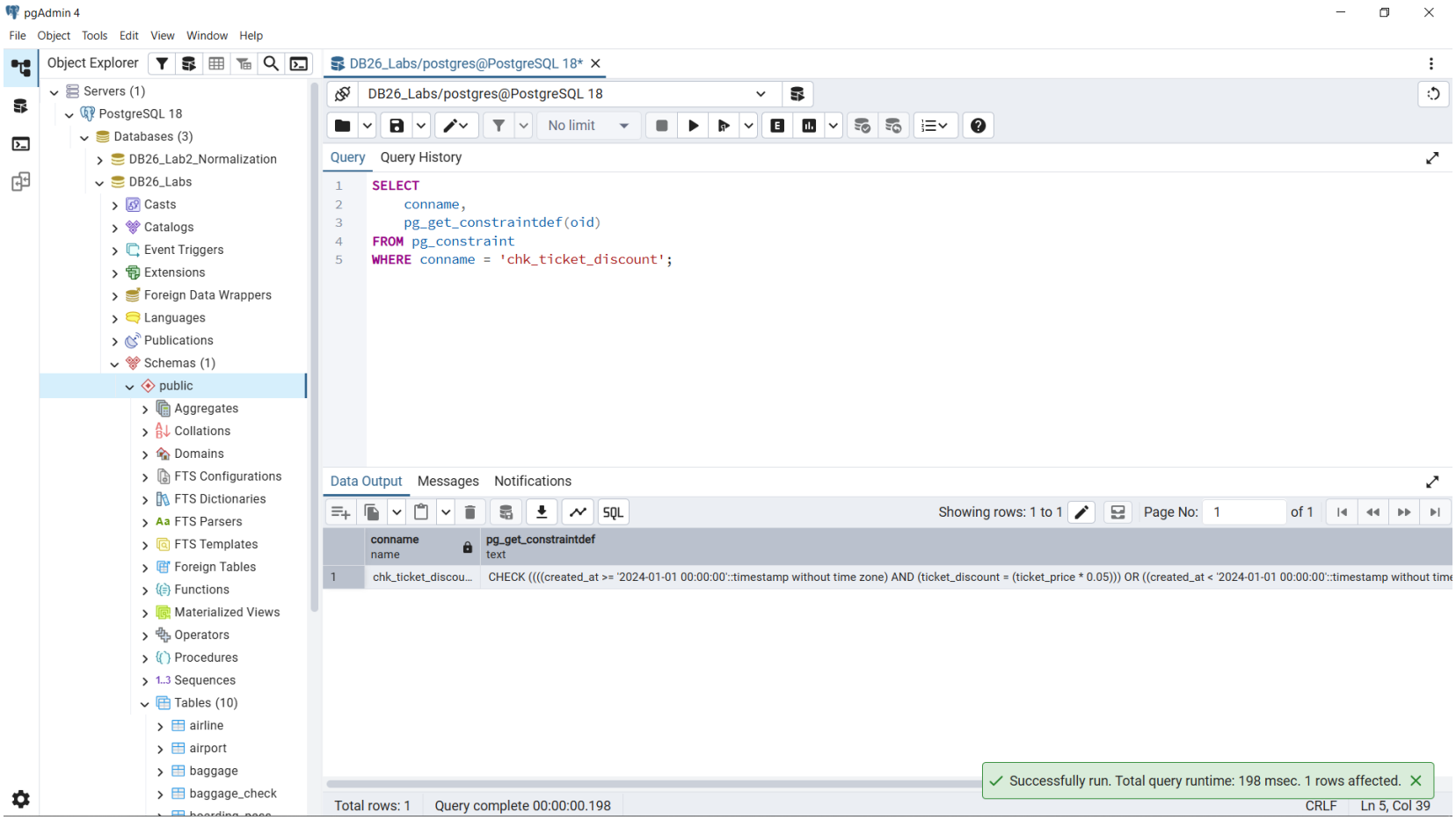
UPDATE Booking
SET ticket_discount = CASE
  WHEN created_at >= '2024-01-01' THEN ticket_price * 0.05
  ELSE ticket_price * 0.10
END;

ALTER TABLE Booking
ADD CONSTRAINT chk_ticket_discount
CHECK ((created_at >= '2024-01-01' AND ticket_discount = ticket_price * 0.05)
    OR (created_at < '2024-01-01' AND ticket_discount = ticket_price * 0.10));
```




pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT
booking_id,
ticket_price,
created_at,
ticket_discount
FROM Booking
LIMIT 10;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 10
Page No: 1
of 1
booking_id [PK] integer
ticket_price numeric (7,2)
created_at timestamp without time zone
ticket_discount numeric (7,2)
1 6 29900.00 2025-01-19 00:00:00 1495.00
2 7 31050.00 2025-01-22 00:00:00 1552.50
3 8 32200.00 2025-01-25 00:00:00 1610.00
4 9 33350.00 2025-01-28 00:00:00 1667.50
5 10 34500.00 2025-01-31 00:00:00 1725.00
6 11 35650.00 2025-02-03 00:00:00 1782.50
7 12 36800.00 2025-02-06 00:00:00 1840.00
8 13 37950.00 2025-02-09 00:00:00 1897.50
9 14 39100.00 2025-02-12 00:00:00 1955.00
10 15 40250.00 2025-02-15 00:00:00 2012.50
Successfully run. Total query runtime: 211 msec. 10 rows affected.
Total rows: 10
Query complete 00:00:00.211
CRLF
Ln 7, Col 10

Task 8 - verification

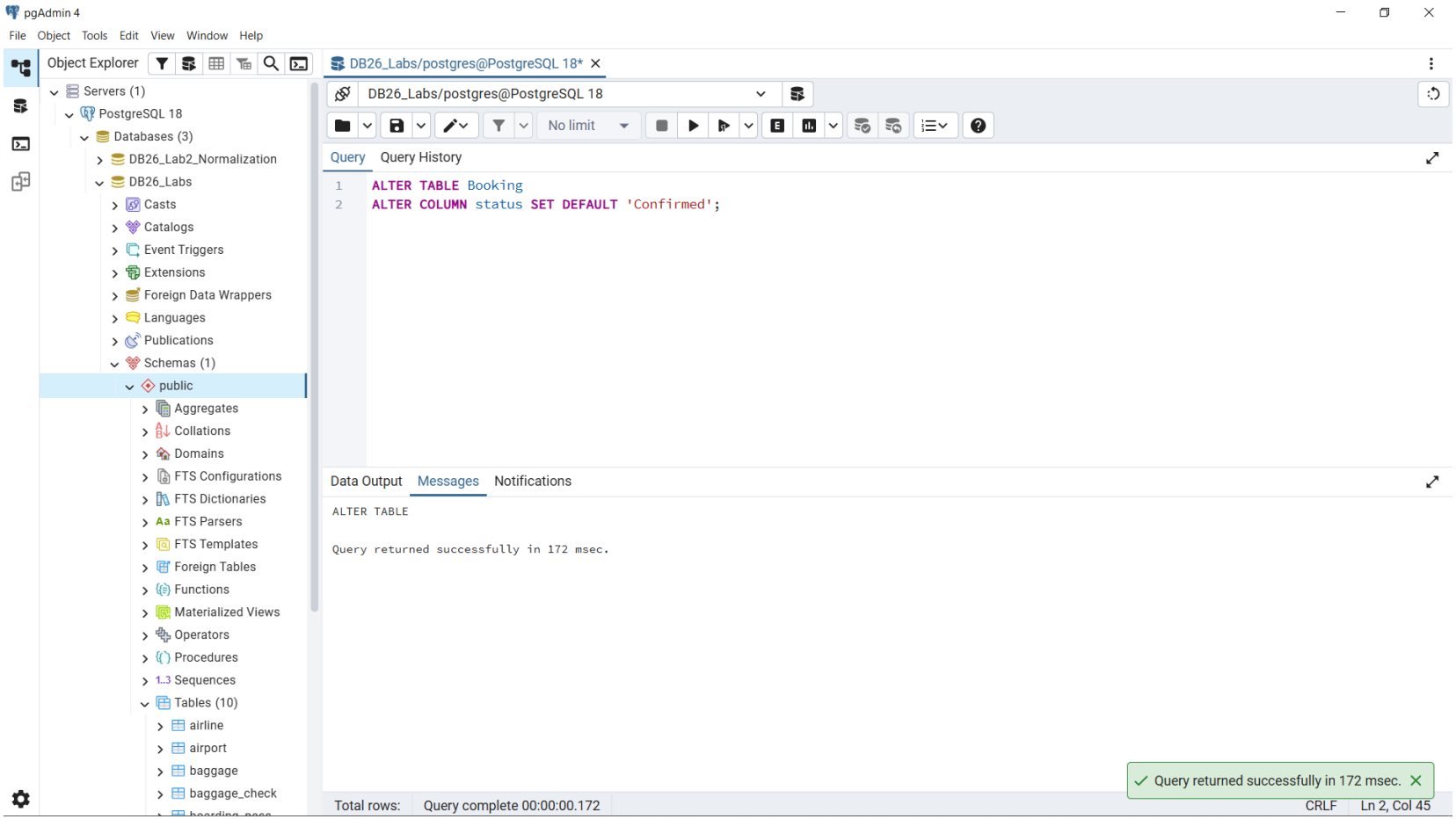


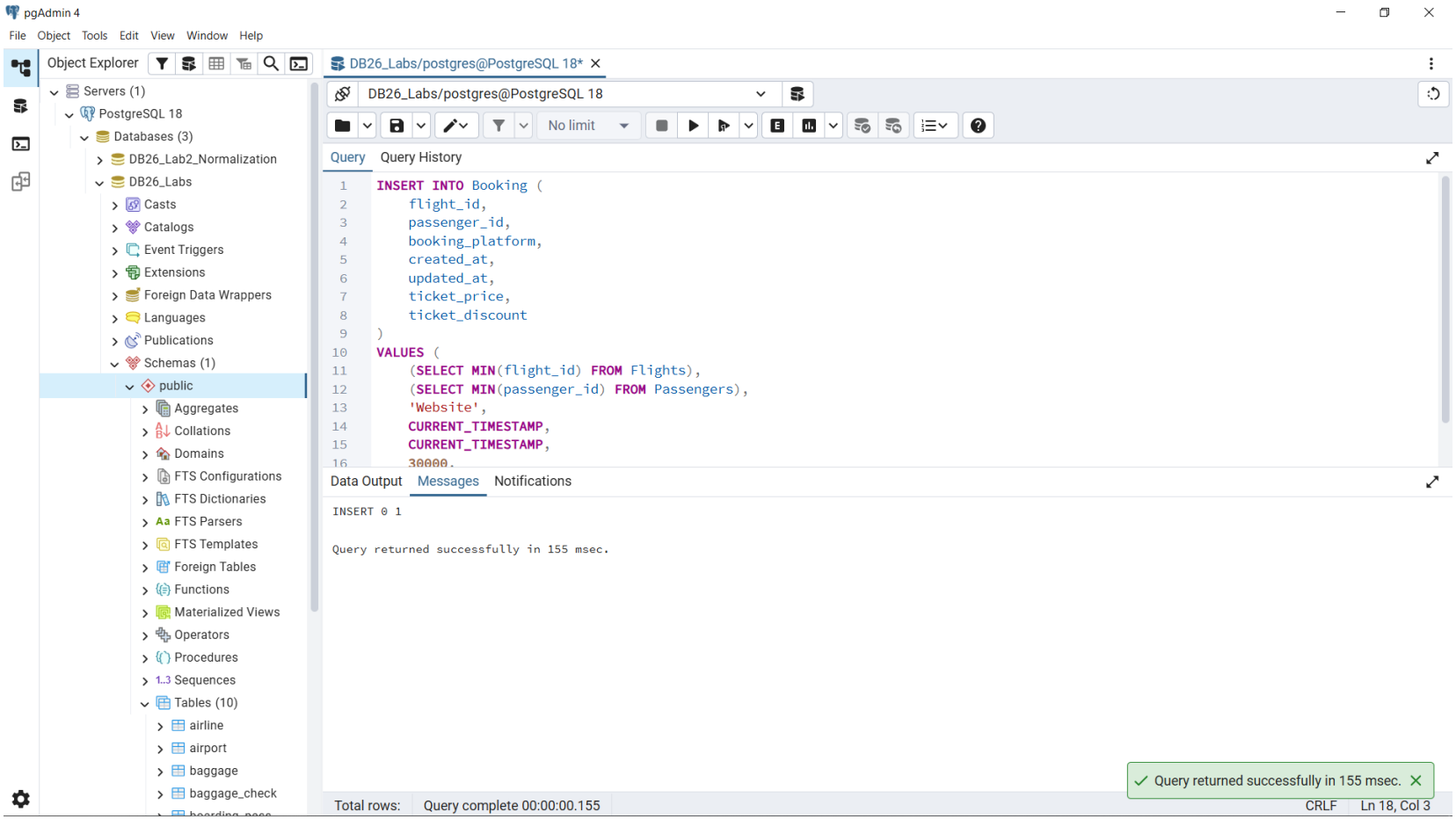
Task 9

Set the default booking status to 'Confirmed' and insert a booking without explicitly specifying status.

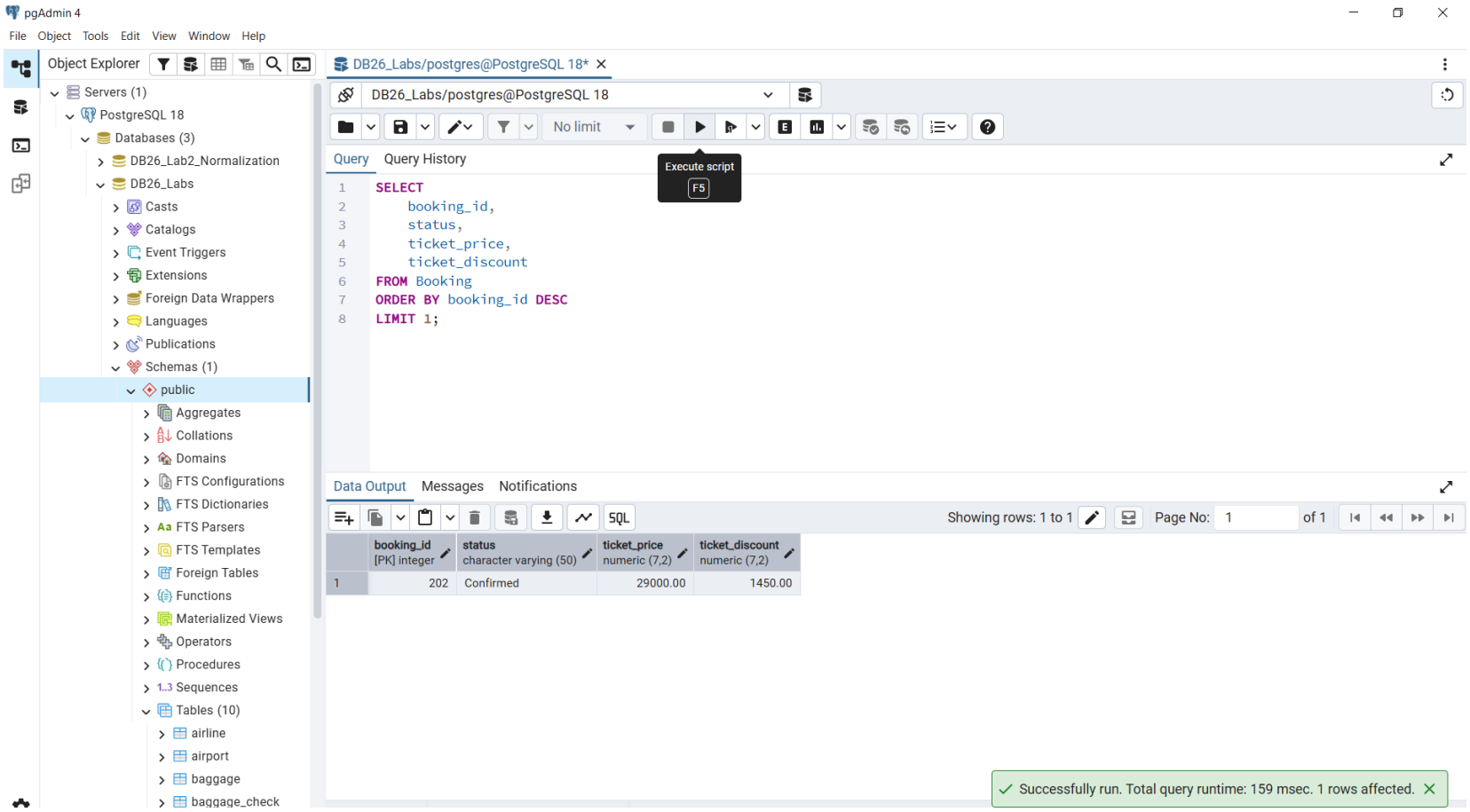
```
ALTER TABLE Booking
ALTER COLUMN status SET DEFAULT 'Confirmed';

INSERT INTO Booking (flight_id, passenger_id, booking_platform, created_at, updated_at, ticket_price, ticket_discount)
VALUES ((SELECT MIN(flight_id) FROM Flights),
    (SELECT MIN(passenger_id) FROM Passengers),
    'Website', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 30000, 1500);
```




pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Query
Query History
INSERT INTO Booking (
flight_id,
passenger_id,
booking_platform,
created_at,
updated_at,
ticket_price,
ticket_discount
)
VALUES (
(SELECT MIN(flight_id) FROM Flights),
(SELECT MIN(passenger_id) FROM Passengers),
'Website',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP,
30000,
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 155 msec.
Query returned successfully in 155 msec.
Total rows:
Query complete 00:00:00.155
CRLF
Ln 18, Col 3

Task 9 - verification

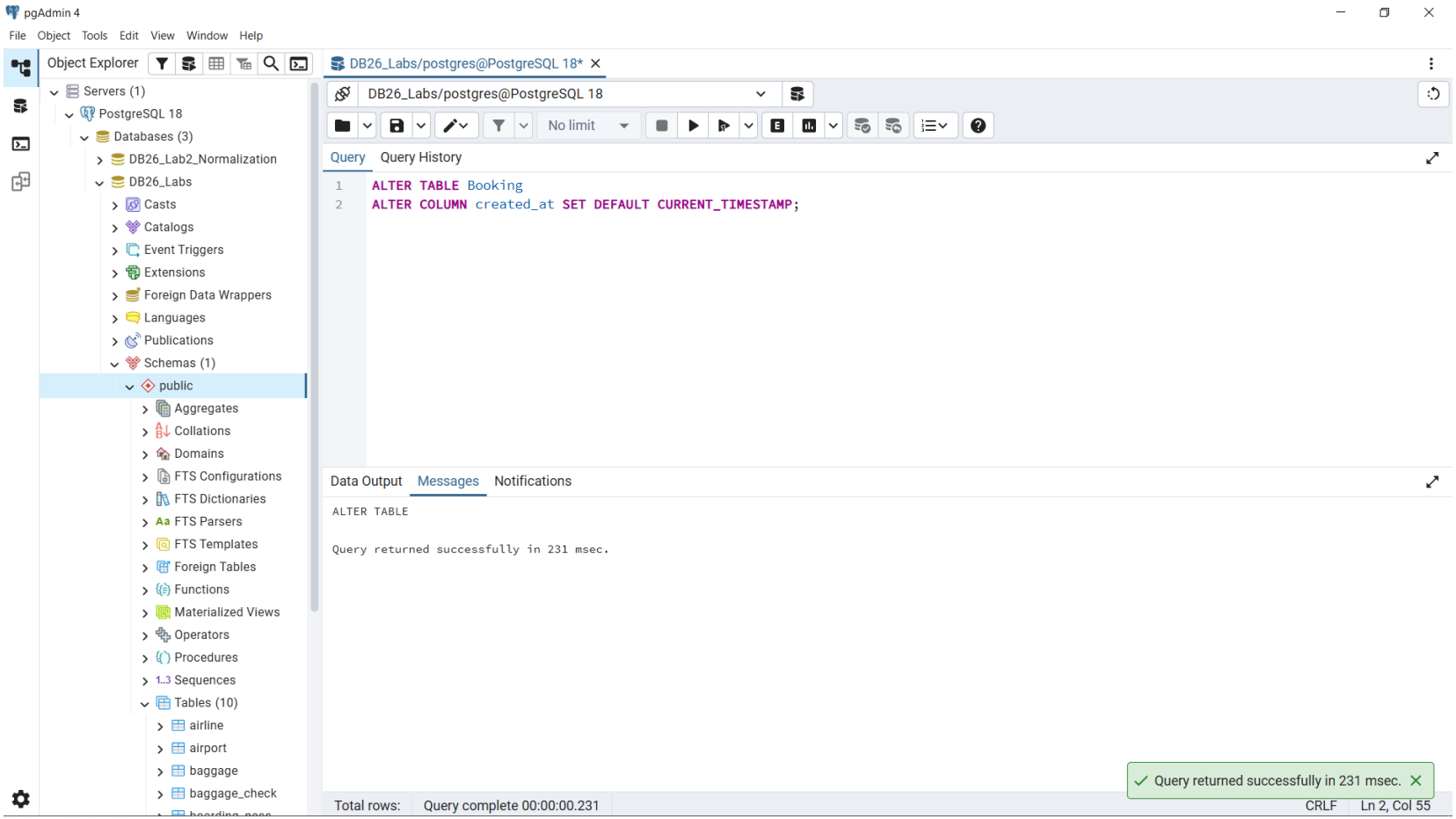


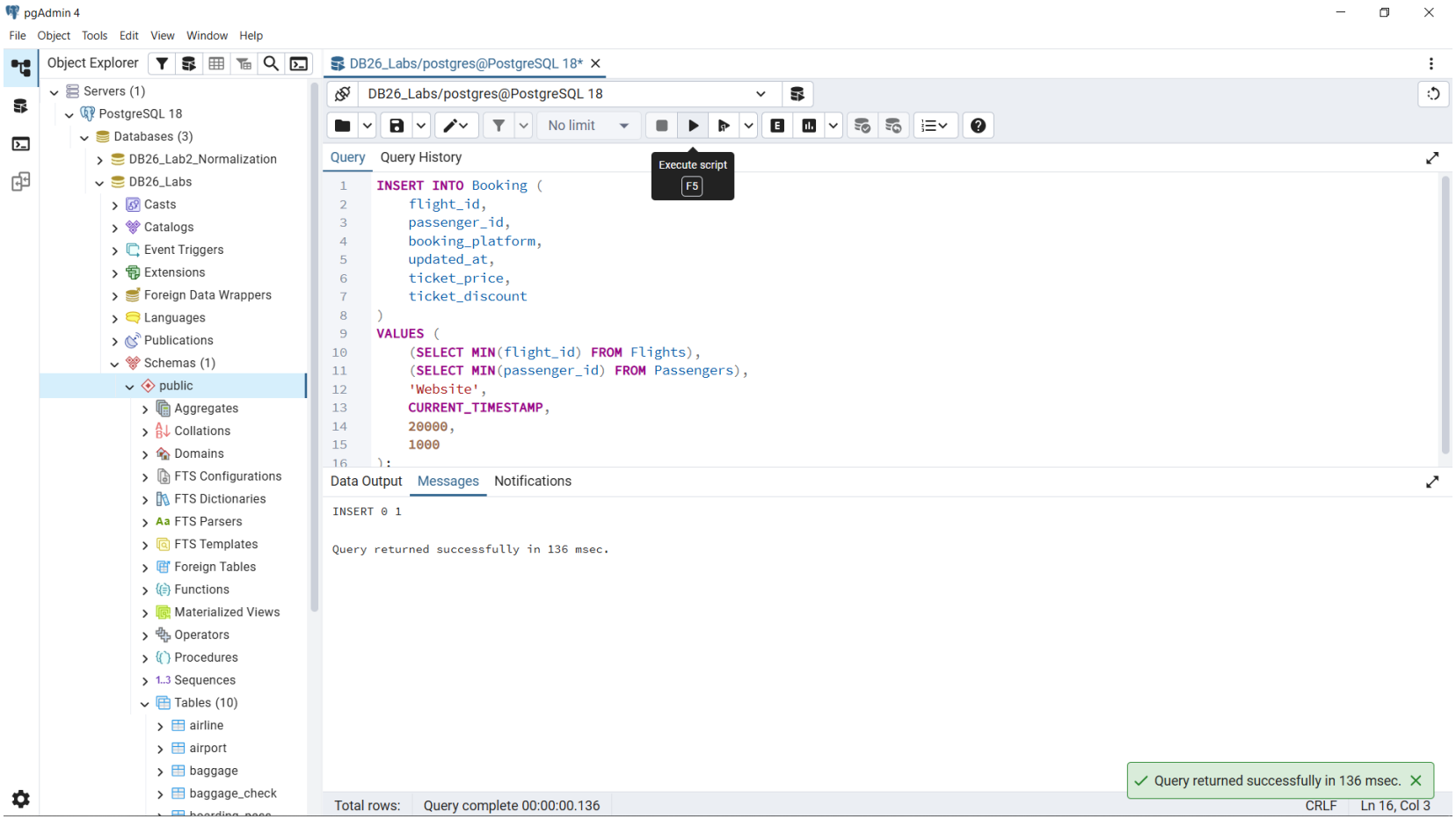
Task 10

Set a suitable date column to the current timestamp automatically when a new row is inserted. Booking.created_at is used.

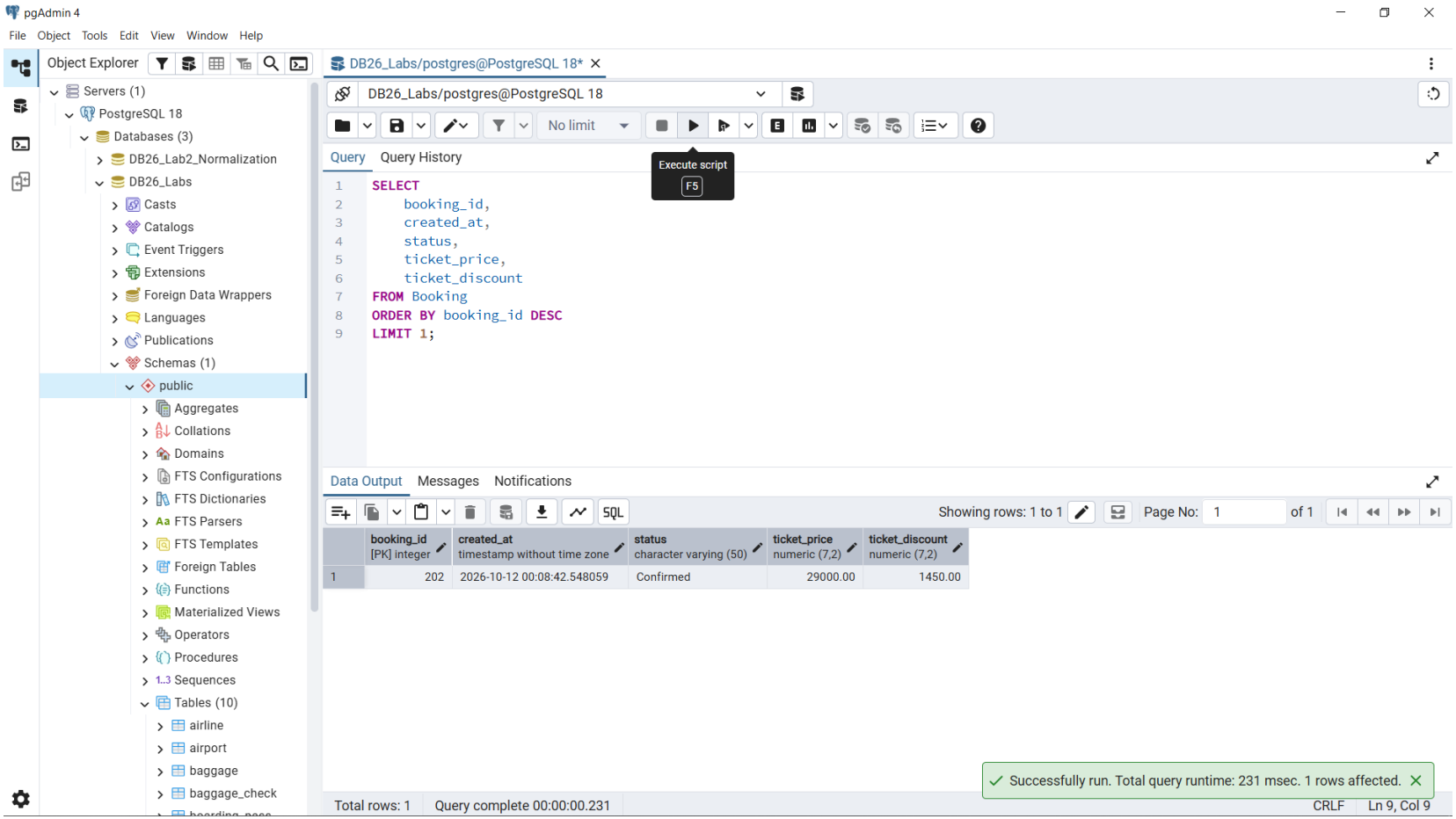
```
ALTER TABLE Booking
ALTER COLUMN created_at SET DEFAULT CURRENT_TIMESTAMP;

INSERT INTO Booking (flight_id, passenger_id, booking_platform, updated_at, ticket_price, ticket_discount)
VALUES ((SELECT MIN(flight_id) FROM Flights),
    (SELECT MIN(passenger_id) FROM Passengers),
    'Website', CURRENT_TIMESTAMP, 20000, 1000);
```




pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (1)
PostgreSQL 18
Databases (3)
DB26_Lab2_Normalization
DB26_Labs
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (10)
airline
airport
baggage
baggage_check
DB26_Labs/postgres@PostgreSQL 18*
DB26_Labs/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
INSERT INTO Booking (
flight_id,
passenger_id,
booking_platform,
updated_at,
ticket_price,
ticket_discount
)
VALUES (
(SELECT MIN(flight_id) FROM Flights),
(SELECT MIN(passenger_id) FROM Passengers),
'Website',
CURRENT_TIMESTAMP,
20000,
1000
);
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 136 msec.
Query returned successfully in 136 msec.
Total rows:
Query complete 00:00:00.136
CRLF
Ln 16, Col 3

Task 10 - verification



Conclusion

All ten tasks of Laboratory Work #5 were completed in PostgreSQL using the DB26_Labs database. The work introduced CHECK constraints for age, ticket price, baggage weight, airport-name length, gender, citizenship and ticket discounts; UNIQUE constraints were added to suitable columns; and DEFAULT values were configured for booking status and creation time. The screenshots confirm successful execution and verification of the implemented database rules in pgAdmin 4.